(ii) Transcendental functions

A function which is not algebraic is called transcendental function.

Examples: $\sin x$, $\sin^{-1} x$, e^x, $\log_a x$ are transcendental functions.

(a) $\sin x$, $\tan 2x$, ... are trigonometric functions.

(b) $sin^{-1} x$, $cos^{-1} x$, ... are inverse trigonometric functions.

(c) e^x, 2^x, x^x, ... are exponential functions.

(d) $log_a x$, $log_e (sin\ x)$, ... are logarithmic functions.

5.1.8 Even and odd functions

A function $f(x)$ is said to be an even function of x, if $f(-x) = f(x)$.

A function $f(x)$ is said to be an odd function of x, if $f(-x) = -f(x)$.

Examples:

$f(x) = x^2$ and $f(x) = cos\ x$ are even functions.

$f(x) = x^3$ *and* $f(x) = sin\ x$ are odd functions.

NOTE

$f(x) = x^3 + 5$ is neither even nor odd function

5.1.9 Explicit and implicit functions

A function in which the dependent variable is expressed explicitly in terms of some independent variables is known as **explicit function**.

Examples: $y = x^2 + 3$ and $y = e^x + e^{-x}$ are explicit functions of x.

If two variables x and y are connected by the relation or function $f(x, y) = 0$ and none of the variable is directly expressed in terms of the other, then the function is called an **implicit function**.

Example: $x^3 + y^3 - xy = 0$ is an implicit function.

5.1.10 Constant function

If k is a fixed real number then the function $f(x)$ given by $f(x) = k$ for all $x \in R$ is called a constant function.

Examples: $y = 3$, $f(x) = -5$ are constant functions.

DO YOU KNOW?

Graph of a constant function $y = f(x)$ is a straight line parallel to x-axis

5.1.11 Identity function

A function that associates each real number to itself is called the identity function and is denoted by I.

i.e. A function defined on R by $f(x) = x$ for all $x \in R$ is an identity function.

Example: The set of ordered pairs $\{(1, 1), (2, 2), (3, 3)\}$ defined by $f : A \to A$ where $A=\{1, 2, 3\}$ is an identity function.

DO YOU KNOW?

Graph of an identity function on R is a straight line passing through the origin and makes an angle of 45° with positive direction of x-axis

5.1.12 Modulus function

A function $f(x)$ defined by $f(x) = |x|$, where $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$ is called the modulus function.

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